MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH & STUDIES

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

Data Structure and Algorithm

Academic Session: 2026–2027

SUBMITTED TO

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Course: MCA

Semester: I

C: > Users > asus > Desktop > DSA_python > lab1_array > insertion.py > ...

```python
size = int(input("Enter the size of the list: "))
my_list=[]
for i in range(size):
    x= int(input("Enter element: "))
    my_list.append(x)
print(my_list)
pos = int(input("Enter the postion of the element: "))
element =  int(input("Enter the element: "))

for i in range(size,pos,-1):
    my_list[i]=my_list[i-1]

my_list.insert(pos,element)
print(my_list)
size=size+1

```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab1_array/insertion.py
Enter element: 5
Enter element: 6
Enter element: 7
Enter element: 8
Enter element: 9
[5, 6, 7, 8, 9]
Enter the postion of the element: 6
Enter the element: 8
[5, 6, 7, 8, 9, 8]
PS C:\Users\asus>
```

Users > asus > Desktop > DSA_python > lab1_array > deletion.py > size

```python
size = int(input("Enter the size of the list: "))
my_list=[]
for i in range(size):
    x= int(input("Enter element: "))
    my_list.append(x)
print(my_list)

pos = int(input("Enter the postion of the element: "))


for i in range(pos,size-1):
    my_list[i]=my_list[i+1]

my_list.pop()
size=size-1
print(my_list)
```

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab1_array/deletion.py
Enter the size of the list: 5
Enter element: 3
Enter element: 4
Enter element: 5
Enter element: 6
Enter element: 7
[3, 4, 5, 6, 7]
Enter the postion of the element: 5
[3, 4, 5, 6]
PS C:\Users\asus>
```

```python
rows1= int(input("Enter the rows of the list: "))
column1=int(input("Enter the column of the list: "))
matrix1=[]
for i in range(rows1):
    row=[]
    for j in range(column1):
        x= int(input("Enter element: "))
        row.append(x)
    matrix1.append(row)

print(matrix1)

diagonal_sum=0
non_diagonal_sum=0
for i in range(rows1):
    row=[]
    for j in range(column1):
        if(i==j):
            diagonal_sum+=matrix1[i][j]
        else:

            non_diagonal_sum+=matrix1[i][j]
print("Sum of Diagonal elements: " ,diagonal_sum)
print("Sum of Non Diagonal elements: " ,non_diagonal_sum)
```

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab1_array/diagonal_sum.py
Enter element: 8
Enter element: 9
Enter element: 7
Enter element: 4
Enter element: 1
[[2, 3, 5], [6, 8, 9], [7, 4, 1]]
Sum of Diagonal elements:  11
Sum of Non Diagonal elements:  34
```

Users > asus > Desktop > DSA_python > lab1_array > matrix_product.py > rows1

```python
rows1= int(input("Enter the rows of the list: "))
column1=int(input("Enter the column of the list: "))
matrix1=[]
for i in range(rows1):
    row=[]
    for j in range(column1):
        x= int(input("Enter element: "))
        row.append(x)
    matrix1.append(row)

print(matrix1)
rows2= int(input("Enter the rows of the list: "))
column2=int(input("Enter the column of the list: "))
matrix2=[]
for i in range(rows2):
    row=[]
    for j in range(column2):
        x= int(input("Enter element: "))
        row.append(x)
    matrix2.append(row)

print(matrix2)

matrix3=[]
for i in range(rows1):
    row=[]
    for j in range(column2):

        row.append(0)
    matrix3.append(row)
for i in range(rows1):
        for j in range(column2):
            for k in range(column1):
                matrix3[i][j] += matrix1[i][k] * matrix2[k][j]

print(matrix3)
```

```
Enter the rows of the list: 3
Enter the column of the list: 3
Enter element: 3
Enter element: 4
Enter element: 3
Enter element: 2
Enter element: 6
Enter element: 5
Enter element: 7
Enter element: 2
Enter element: 1
[[3, 4, 3], [2, 6, 5], [7, 2, 1]]
Enter the rows of the list: 3
Enter the column of the list: 3
Enter element: 1
Enter element: 2
Enter element: 3
Enter element: 4
Enter element: 5
Enter element: 6
Enter element: 7
Enter element: 8
Enter element: 9
[[1, 2, 3], [4, 5, 6], [7, 8, 9]]
[[40, 50, 60], [61, 74, 87], [22, 32, 42]]
PS C:\Users\asus>
```

```python
rows1= int(input("Enter the rows of the list: "))
column1=int(input("Enter the column of the list: "))
matrix1=[]
for i in range(rows1):
    row=[]
    for j in range(column1):
        x= int(input("Enter element: "))
        row.append(x)
    matrix1.append(row)

print(matrix1)
for i in range(rows1):
    row=[]
    total_marks =0
    for j in range(column1):
        total_marks+= matrix1[i][j]
    print(f"Percentage of student{i}: ",(total_marks/(column1*100)*100))
```

(variable) rows1: int

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab1_array/marks_percentage.py
Enter element: 96
Enter element: 69
Enter element: 95
[[78, 78, 89], [89, 89, 65], [99, 99, 100], [36, 63, 75], [96, 69, 95]]
Percentage of student0:  81.66666666666667
Percentage of student1:  81.0
Percentage of student2:  99.33333333333333
Percentage of student3:  57.99999999999999
Percentage of student4:  86.66666666666667
PS C:\Users\asus>
```

```python
stack = []
top = -1
size = 5

while True:
    print("\n--- STACK OPERATIONS ---")
    print("1. Push")
    print("2. Pop")
    print("3. Display")
    print("4. Peek")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if top == size - 1:
            print("Stack Overflow")

        else:
            element = int(input("Enter element to push: "))
            stack = stack + [element]
            top = top + 1
            print("Element pushed successfully")

    elif choice == 2:
        if top == -1:
            print("Stack Underflow")

        else:
            print("Deleted element:", stack[top])
            stack = stack[:top]
            top = top - 1

    elif choice == 3:
        if top == -1:
            print("Stack is empty")

        else:
            print("Stack elements (Top to Bottom):")
            i = top

            while i >= 0:
                print(stack[i])
                i = i - 1

    elif choice == 4:
        if top == -1:
            print("Stack is empty")

        else:
            print("Top element:", stack[top])

    elif choice == 5:
        print("Program terminated")
        break

    else:
        print("Invalid choice")
```

```
--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element to push: 36
Element pushed successfully

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element to push: 32
Element pushed successfully

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 15
Invalid choice

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 1
Enter element to push: 6
Element pushed successfully
```

```
Element pushed successfully

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 2
Deleted element: 6

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 3
Stack elements (Top to Bottom):
32
36

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 4
Top element: 32

--- STACK OPERATIONS ---
1. Push
2. Pop
3. Display
4. Peek
5. Exit
Enter your choice: 5
Program terminated
```

```python

word = input("Enter the word: ")
stack=[0]*len(word)
top = -1
for s in word:
    top+=1
    stack[top]=s
rev =""
while(top!=-1):
    rev += stack[top]
    top-=1
print(rev)

```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/Lab2_stack/reverse_stack.py
Enter the word: Arjun Singh Kushwah
hawhsuK hgniS nujrA
PS C:\Users\asus>
```

```python
equation = input("Enter the equation")
stack=[]
top=-1

for ch in equation:
    if(ch=="("):
        stack.append(ch)
        top+=1
        print(stack)
    elif(ch==")"):
        ele=stack.pop()
        print(ele)
        top-=1
        print(stack)
```

```
Enter the equation ((a+b)^2-(c-d)^2)/sqrt((x+y)*(x-y))
['(']
['(', '(']
(
['(']
['(', '(']
(
['(']
(
[]
['(']
['(', '(']
(
['(']
['(', '(']
(
['(']
(
[]
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/Lab2_stack/par
Enter the equation
```

```python
size = int(input("Enter the size of Queue: "))
queue = [0] * size
front = -1
rear = -1


while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if rear == size - 1:
            print("Queue Overflow")
        else:
            element = int(input("Enter element: "))

            if front == -1:
                front = 0

            rear = rear + 1
            queue[rear] = element
            print("Element inserted:", element)

    elif choice == 2:
        if front == -1 or front > rear:
            print("Queue Underflow")
        else:
            print("Deleted element:", queue[front])
            front = front + 1

            if front > rear:
                front = -1
                rear = -1

    elif choice == 3:
        if front == -1:
            print("Queue is empty")
        else:
            print("Queue elements:")
            for i in range(front, rear + 1):
                print(queue[i], end=" ")
            print()

    elif choice == 4:
        print("Program terminated")
        break

    else:
        print("Invalid choice")
```

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab3_queu

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 5
Element inserted: 5

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 2
Element inserted: 2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted element: 5

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements:
2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Program terminated
```

Users > asus > Desktop > DSA_python > lab3_queue > circular.py > size

```python
size = int(input("Enter the size of Queue:"))
queue = [0] * size
front = -1
rear = -1


while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if (rear + 1) % size == front:
            print("Circular Queue Overflow")
        else:
            element = int(input("Enter element: "))

            if front == -1:
                front = 0
                rear = 0
            else:
                rear = (rear + 1) % size

            queue[rear] = element
            print("Element inserted:", element)

    elif choice == 2:
        if front == -1:
            print("Circular Queue Underflow")
        else:
            print("Deleted element:", queue[front])

            if front == rear:
                front = -1
                rear = -1
            else:
                front = (front + 1) % size

    elif choice == 3:
        if front == -1:
            print("Circular Queue is empty")
        else:
            print("Queue elements:")

            i = front
            while True:
                print(queue[i], end=" ")

                if i == rear:
                    break

                i = (i + 1) % size

            print()

    elif choice == 4:
        print("Program terminated")
        break

    else:
        print("Invalid choice")
```

```
PS C:\Users\asus> & c:/python314/python.exe c:/Users/asus/Desktop/DSA_python/lab3_queue/circular.py
Enter the size of Queue:5

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 2
Element inserted: 2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 3
Element inserted: 3

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted element: 2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements:
3

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Program terminated
```